

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A	A,B,D	A,C	A,B,C,D	A,B,D	A,C	B,C	B,C	A,B
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	2	5	9	1	2	2	3	3		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		P,R	S,T	P,Q,R	P,Q,R		P	Q	R	S	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,B,D	A,B,C,D	B,C	A,B,C,D	A,C,D	B,D	A,C	A,D	A,D
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	3	5	4	2	5	9	6	4		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		P,R,T	P,S	Q,R,T	Q,R,S		P,S,T	R,S,T	P,R,S	Q,S	

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B,C	B	A,B	A,C	B,C	A,D	A,C	A,C,D	B,C
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	9	1	0	3	0	2	2	0		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		P,T	P,S,T	P,T	Q,S,T		P,Q,R,S,T	P,Q,S	Q	Q	

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,D	A,C	A,B,C	B,D	A,B,D	A,C,D	A,D	A,B,D	A,C,D	A,C
	Q.	11	12								
	A.	B,C	B,C								
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	3	3	2	6	2	5	5	1		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	A,B,C,D	A,B,C	B,C,D	A,C	A,C,D	B,C	A,B,C,D	B,D	A,C,D
	Q.	11	12								
	A.	A,B,C	A,C								
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	2	4	9	6	2	1	5	2		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,C,D	B,D	A,B,D	B,C	B,C,D	B	B,C	A,D	D
	Q.	11	12								
	A.	A	A,B								
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	4	4	2	2	4	2	0	6		

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A,C,D)**

$$\text{Sol. } V_A - V_D = \frac{\frac{6}{2} + \frac{12}{6} - \frac{3}{3}}{\frac{1}{2} + \frac{1}{6} + \frac{1}{3}} = \frac{4}{1}$$

For first arm

$$V_A - 6 - x \times 2 = V_B$$

$$V_A - V_B - 6 = x \times 2$$

$$x = -1$$

For second arm

$$V_B - 6y - 12 = V_E$$

$$V_B - V_E - 12 = 6y$$

$$y = -\frac{8}{6} = -\frac{4}{3}$$

For third arm

$$V_{A+B} - 32 = V_B$$

$$\frac{7}{3} = 2$$

2. **Ans. (A)**

$$\text{Sol. } x + 4 = 2 + y$$

$$x + 2 = y$$

3. **Ans. (A,B,D)**

$$\text{Sol. } \frac{2\pi}{\tau} = \sqrt{\frac{G(M_1 + M_2)}{(R_1 + R_2)^3}}$$

Total energy of system is conserved otherwise system will not be able to maintain motion.

Areal velocity is conserve that's why it is repetition its motion.

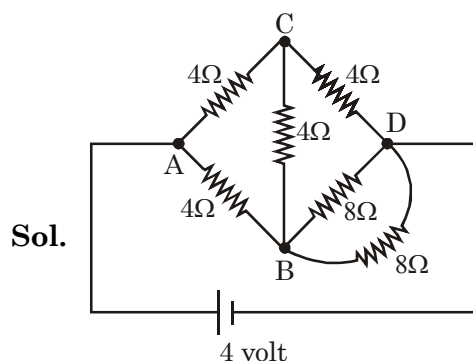
4. **Ans. (A,C)**

$$\text{Sol. } \rho_1 V_1 = \rho_2 V_2 \left[V = \frac{4}{3} \pi R^2 \right]$$

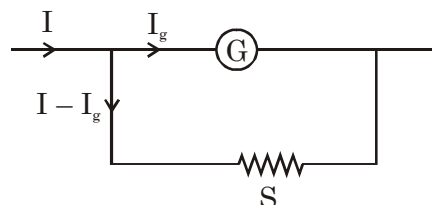
$$\frac{R_1^3}{R_2^2} = \frac{\rho_2}{\rho_1} \quad \dots(i)$$

$$g = \frac{GM}{R^2} \text{ and } V = -\frac{GM}{R}$$

$$\frac{g_1}{g_2} \Rightarrow \frac{R_2^2}{R_1^2} \text{ and } \frac{V_1}{V_2} \Rightarrow \frac{R_2}{R_1}$$

5. **Ans. (A,B,C,D)**

It satisfies Wheatstone bridge so $V_B = V_C$ and no current in wire B to C. $R_{eq} = 4\Omega$
 So current through battery is 1A and it divides in proportions.

6. **Ans. (A,B,D)****Sol.** For ammeter

$$(I_g)G = (I - I_g)S$$

$$I_g = 10^{-4} \times 200$$

$$G = 198$$

For voltmeter



$$I_g(R + G) = V$$

7. Ans. (A,C)

Sol. According to Wein's Law,

$$T \propto \frac{1}{\lambda}$$

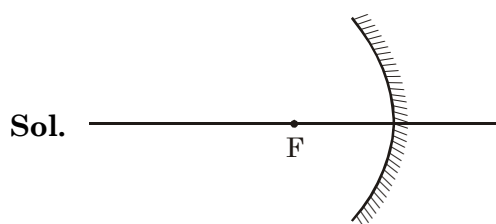
$$T \propto \nu$$

It T is double, corresponding ν also doubles.

Total energy $\propto T^4$

So total energy will becomes 16 times.

8. Ans. (B,C)



$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$v_i = m^2 v_o$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{u-f}{fu}$$

$$v = \frac{fu}{u-f}$$

$$\frac{v}{u} = \frac{f}{u-f} = m$$

where $\frac{v}{u} = m$

u decreases so m increases as speed of image as well as height of image.

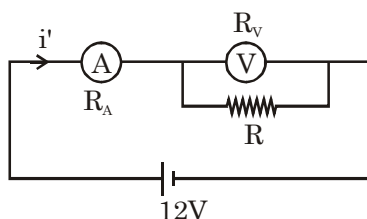
9. Ans. (B,C)

Sol. Initial current = $i = \frac{12}{R_A + R_V}$

& P.d. across voltmeter

$$= (V) = i R_{eq} = \frac{(12) R_V}{(R_A + R_V)}$$

When unknown resistance 'R' is connected in parallel to voltmeter



$$\text{Final current} = i' = \frac{12}{R_A + \left(\frac{R_V R}{R_V + R} \right)}$$

$$\text{p.d. across voltmeter} = V' = i' \times \frac{R_V R}{(R_V + R)}$$

$$i' = 3i \text{ \& } V' = \frac{V}{3}$$

$$\frac{12}{R_A + \frac{R_V R}{R_V + R}} = 3 \times \frac{12}{(R_A + R_V)}$$

$$\& \frac{12}{R_A + \frac{R_V R}{R_V + R}} \cdot \frac{(R_V R)}{(R_V + R)} = \frac{4R_V}{(R_A + R_V)}$$

$$\text{On solving : } R_V = 8R, R_A = \frac{8R}{3}$$

Also $V = 9V$ i.e. reading of voltmeter before connecting unknown resistance

10. Ans. (A,B)

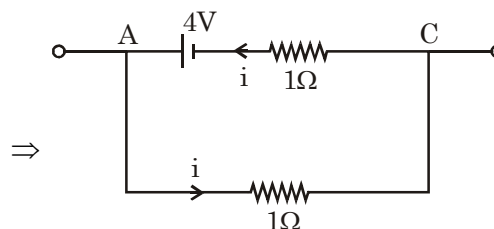
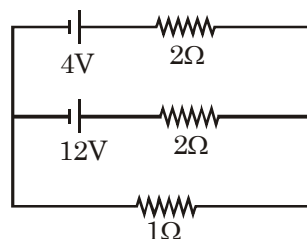
Sol. When S_1 is closed, null point cannot be obtained because secondary cell of 4V is in same polarity and p.d. cannot be compared. When S_2 is closed, null point length

$$= \frac{12V}{\text{Potential gradient}}$$

$$= \frac{12V}{8V/m} = 1.5m$$

When S_1 and S_2 are both closed, p.d. across terminal = 4V with A at high potential, So null point can be obtained.

When S_1, S_2 & S_3 are closed



$$\Rightarrow V_A - V_C = 2V$$

$$i = \frac{4}{2} = 2A$$

∴ Balance point obtained at

$$\ell = \frac{2V}{\text{Point gradient}} = \frac{2V}{8V/m} = 0.25m$$

$$= 25 \text{ cm from A.}$$

SECTION-III

1. **Ans. 2**

Sol. ∴ $I = \frac{V}{R} = \frac{200}{80} = 2.5A$

& $P = \frac{V^2}{R} = \frac{(200)^2}{80} = 500 \text{ watt}$

When the wire is divided into two equal parts and connected in parallel we get maximum power

∴ Resistance of each part = 40Ω

∴ $I_{\max} = 5A$

⇒ $P_{\max} = \frac{V^2}{R_{\text{eq}}} = \frac{(200)^2}{(20)} = 2000W = 2kw.$

2. **Ans. 5**

Sol. $\frac{F/A}{\Delta\ell/\ell} = y$

$F = \left(\frac{YA}{\ell}\right)\Delta\ell$

$T_1 = 2\pi\sqrt{\frac{m\ell}{yA}} \quad \omega = \sqrt{\frac{yA}{m\ell}}$

$\tau = \tau\theta$

$\alpha = \left(\frac{C}{\frac{2}{5}MR^2}\right)\theta$

$\omega = \sqrt{\frac{5C}{2MR^2}} \quad T_2 = 2\pi\sqrt{\frac{2M}{5C}}R$

$\frac{T_1}{T_2} = \frac{1}{R}\sqrt{\frac{5\ell C}{yA2}} = \frac{1}{R}\sqrt{\frac{5C\ell}{2yA}}$

3. **Ans. 9**

Sol. $\Delta P \Rightarrow -\beta\left(\frac{\Delta V}{V}\right)$ and $\Delta V \Rightarrow Vy\Delta T$

$\Delta P \Rightarrow \beta y\Delta T \quad y = 3\alpha$

on putting values $\Delta P \approx 9.24 \times 10^8$

4. **Ans. 1**

Sol. Let final temperature be $T^\circ\text{C}.$

Heat absorbed by ice to reach $T^\circ\text{C}.$

$(m_{\text{ice}})(S_{\text{ice}})(0 - (-10^\circ)) + (m_{\text{ice}})(L_f)$
 $+ (m_{\text{ice}})(S_w)(T - 0)$

$m_{\text{ice}} = 10\text{gm}, S_{\text{ice}} = 0.5 \text{ cal/gm}^\circ\text{C}$

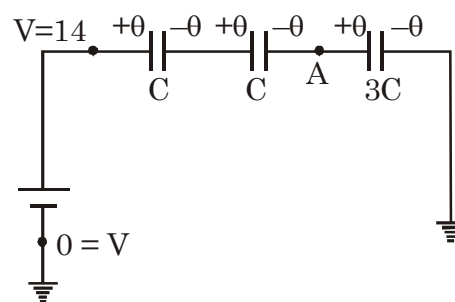
Heat released by water while coming down to $T^\circ\text{C}.$

$(m_w)(S_w)(30 - T)$

Heat absorbed = Heat released.

on putting values $T = \frac{5}{4} = 1.25^\circ\text{C}$

5. **Ans. 2**



Sol.

$+14 - \frac{\theta}{C} - \frac{\theta}{C} - \frac{\theta}{3C} = 0 \quad \dots(i)$

$\frac{\theta}{C} = 6$

$V_A - \frac{\theta}{3C} = 0 \quad \dots(ii)$

∴ $V_A = 2.$

6. **Ans. 2**

Sol. Gravitational field at C is $= \frac{GM/8}{(R/2)^2} = \frac{GM}{2R^2}$

Gravitational field constant in cavity.

for time

$S = at + \frac{1}{2}at^2$

$\frac{R}{2} = 0 + \frac{1}{2}\left(\frac{GM}{2R^2}\right)t^2$

$t = \sqrt{\frac{2R^3}{GM}} \quad a = 2$

7. **Ans. 3**

Sol. Resistance of small element of length dx

⇒ $dR = \frac{\rho dx}{A} = \left[\frac{\rho_0 x}{\ell} \cdot \frac{dx}{A}\right]$

On balancing : $\frac{2}{4} = \frac{R_1}{R_2} = \frac{\int_0^x dR}{\int_x^\ell dR}$

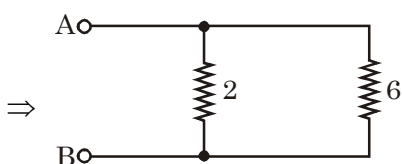
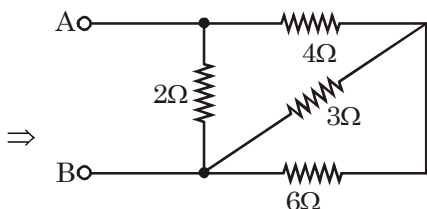
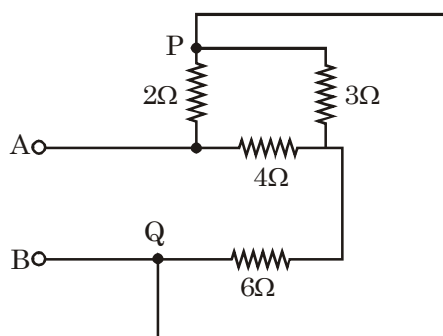
i.e. $\frac{1}{2} = \frac{\int_0^x x dx}{\int_x^\ell x dx} \Rightarrow \frac{1}{2} = \frac{x^2}{\ell^2 - x^2}$

$\therefore \ell^2 - x^2 = 2x^2$

$x = \frac{\ell}{\sqrt{3}}$

8. Ans. 3

Sol. The circuit can be reconstructed as :



i.e. P & Q are at same potential & so can be connected.

$\Rightarrow R_{AB} = \frac{6 \times 2}{6 + 2} \Omega = \frac{3}{2} \Omega$

SECTION-IV

1. Ans. (A)-P,R; (B)-S,T; (C)-P,Q,R; (D)-P,Q,R

Sol. (A) Cyclic process

Internal energy depends on temperature.
more IE $\Rightarrow$ High temperature in graph
volume expands at higher temperature and

decreases at lower temperature. Overall work done positive

(B) $PM = \rho RT$

$P = \text{constant}$

$\Rightarrow \rho \uparrow T \downarrow$

(C) Even though initial and final volume are same but volume decreases at lower temperature and increases at higher

$\Rightarrow$ Work done = positive

(D) $P = kV \Rightarrow PV^{-1} = k$

for polytropic process when $x = -1$

$\frac{nRT}{V} = kV \Rightarrow nRT = kV^2$

$\Rightarrow$ Volume $\uparrow \Rightarrow T \uparrow$

2. Ans. (A)-P; (B)-Q; (C)-R; (D)-S

Sol. (A) Thermal resistance (R_T)

$= \frac{\ell}{KA} = \frac{3R}{K\pi((2R)^2 - R^2)} = \frac{R}{\pi KR^2} = \frac{1}{\pi KR}$

$\frac{dQ}{dt} = \frac{\Delta T}{R_T} = (T_1 - T_2)\pi RK$

(B) $R_T = \int_R^{2R} \frac{dr}{K(2\pi r)3R} = \frac{1}{6\pi KR} \ln 2;$

$\frac{dQ}{dt} = \frac{\Delta T}{R_T} = \frac{(T_1 - T_2)6\pi KR}{\ln 2}$

(C) $R_T = \int_R^{2R} \frac{dr}{K4\pi r^2} = \frac{1}{4\pi K} \left[\frac{1}{R} - \frac{1}{2R} \right] = \frac{1}{8\pi KR};$

$\frac{dQ}{dt} = 8\pi KR(T_1 - T_2)$

(D) $R_T = \frac{2R}{k\pi R^2} = \frac{2}{\pi KR};$

$\frac{dQ}{dt} = (T_1 - T_2) \frac{\pi KR}{2}$

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**

1. Ans.(A,B,C)
2. Ans.(A,B,D)
3. Ans. (A,B,C,D)
4. Ans.(B,C)

Slow step is rate determining step

$$R = k[\text{NO}]^2[\text{H}_2]$$

5. Ans.(A,B,C,D)

In fcc each atom is surrounded by six octahedral and tetrahedral void.

6. Ans.(A,C,D)

$$1 \text{ curie} = 3.7 \times 10^{10} \text{ Bq}$$

7. Ans.(B,D)

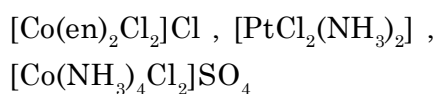
8. Ans.(A,C)

9. Ans.(A,D)

10. Ans.(A,D)

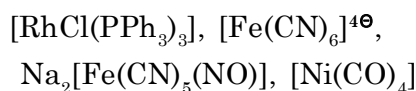
SECTION-III

1. Ans.(3)

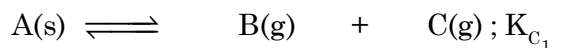


2. Ans.(5)

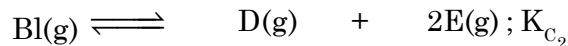
3. Ans.(4)



4. Ans.(20) OMR ANS(2)



$$0.2 \quad \quad \quad 0.04 \quad \quad \quad 0.05$$



$$(0.05 - 0.01) \quad 0.01 \quad \quad \quad 0.02$$

$$K_{C_1} = 0.04 \times 0.05 = 2 \times 10^{-3}$$

$$K_{C_2} = \frac{0.01 \times (0.02)^2}{0.04} = 10^{-4}$$

5. Ans.(1.40) OMR ANS(5)

$$\text{Zero order } t_{1/2} = \frac{[A_0]}{2K}$$

$$\text{first order } t_{1/2} = \frac{\ln 2}{K}$$

$$\frac{[A_0]}{2K} = \frac{\ln 2}{K}$$

$$[A_0] = 2 \times 0.7 = 1.4 \text{ M}$$

6. Ans.(0.09) OMR ANS(9)

7. Ans.(6)

8. Ans.(4)

SECTION-IV

1. Ans. (A)-(P,R,T); (B)-(P,S); (C)-(Q,R,T); (D)-(Q,R,S)

2. Ans.(A)-(P,S,T); (B)-(R,S,T); (C)-(P,R,S); (D)-(Q,S)

PART-3 : MATHEMATICS**SOLUTION****SECTION-I**

1. Ans. (B)

$$\text{Sol. } (1)^\infty = \lim_{x \rightarrow 0} e^{\frac{n}{x^2}} (\cos(mx) - 1)$$

$$= e^{\frac{-(1-\cos(mx))}{(mx)^2} \times m^2 n} = e^{\frac{m^2 n}{2}}$$

2. Ans. (B,C)

Sol. Let centre is $(\alpha, 2\alpha)$ where α is arbitrary constant

$$(x - \alpha)^2 + (y - 2\alpha)^2 = 1$$

$$x^2 + y^2 + (5\alpha^2 - 1) = (2x + 4y)\alpha \quad \dots (1)$$

$$2x + 2yy' = (2 + 4y')\alpha$$

$$x^2 + y^2 + 5\left(\frac{2x + 2yy'}{2 + 4y'}\right)^2 - 1 = \frac{(2x + 2yy')}{(2 + 4y')} (2x + 4y)$$

By solving order = 1
degree = 2

3. Ans. (B)

$$\text{Sol. } \frac{\pi}{2} \int_{-2}^{-1} (5+x)^{\frac{1}{2}} dx = \frac{\pi}{2} \left(\frac{(5+x)^{3/2}}{3/2} \right)_{-2}^{-1} = \frac{\pi}{3} \left(4^{\frac{3}{2}} - 3^{\frac{3}{2}} \right)$$

$$= \frac{\pi}{3} (8 - 3\sqrt{3})$$

4. Ans. (A,B)

$$\text{Sol. } x = e^{\frac{xy}{dx}} \Rightarrow xy \frac{dy}{dx} = \ln x$$

$$y dy = \frac{\ln x}{x} dx \Rightarrow \frac{y^2}{2} = \frac{(\ln x)^2}{2} + C$$

$$y^2 = (\ln x)^2 + C$$

$$y = \pm \ln x$$

5. Ans. (A,C)

$$\begin{aligned}\text{Sol. } \int \frac{D(f(x)g(x))}{1+(f(x)g(x))^2} &= \tan^{-1}(f(x)g(x)) + C \\ &= \frac{\pi}{2} - \cot^{-1}(f(x)g(x)) + C \\ &= -\cot^{-1}(f(x)g(x)) + C\end{aligned}$$

6. Ans. (B,C)

$$\text{Sol. } f''(x) = ax \quad (\text{where } a > 0)$$

$$f'(x) = \frac{ax^2}{2} + b \Rightarrow \frac{a}{2} + b = 0$$

$$f(x) = \frac{ax^3}{6} + bx + C$$

$$\text{By solving } f(x) = \frac{1}{4}(19x^3 - 57x + 34)$$

7. Ans. (A,D)

$$\text{Sol. } I = \int_1^\infty \frac{(x-1)^4 dx}{(x^8 + x^{10})}$$

$$\text{Let } x = \frac{1}{t}, \quad dx = -\frac{dt}{t^2}$$

$$I = \int_1^0 \frac{\left(\frac{1}{t}-1\right)^4 \left(-\frac{dt}{t^2}\right)}{\left(\frac{1}{t^8} + \frac{1}{t^{10}}\right)} = \int_0^1 \frac{(1-t)^4 t^4}{(t^2+1)} dt$$

$$\int_0^1 t^4 \frac{((1+t^2)-2t)^2}{(t^2+1)} dt$$

$$\int_0^1 \left(((1+t^2)t^4) - 4t^5 + \frac{4t^6}{(t^2+1)} \right) dt$$

$$\int_0^1 t^4 dt + t^6 dt - 4t^5 dt + 4 \int_0^1 \frac{(t^6+1)-1}{(t^2+1)} dt$$

8. Ans. (A,C)

$$\text{Sol. (a)} \int_0^1 x^{(m-1)} (1-x)^{(n-1)} dx = \int_0^1 (1-x)^{(m-1)} x^{(n-1)} dx$$

$$(b) \text{ Put } \frac{1}{x} - 1 = t$$

$$\frac{1}{x} = (1+t) \Rightarrow x = \frac{1}{(1+t)} \Rightarrow dx = -\frac{dt}{(1+t)^2}$$

$$-\int_\infty^0 \frac{dt}{(1+t)^2} \frac{1}{(1+t)^{(m-1)}} \times \left(1 - \frac{1}{(1+t)}\right)^{(n-1)}$$

$$\int_0^\infty \frac{dt}{(1+t)^{(m+n)}} t^{(n-1)} = \int_0^\infty \frac{t^{(m-1)} dt}{(1+t)^{(m+n)}}$$

Reason-same replacement for m & n.

$$(c) \text{ Let } x = \sin^2 \theta$$

$$dx = 2\cos\theta \cdot \sin\theta d\theta$$

$$2 \int_0^{\pi/2} (\sin\theta)^{(2m-2)} \cdot (\cos\theta)^{(2n-2)} \cdot \sin\theta \cos\theta d\theta$$

$$2 \int_0^{\pi/2} (\sin\theta)^{(2m-1)} (\cos\theta)^{(2n-1)} d\theta$$

9. Ans. (A,C,D)

$$\text{Sol. } \int_a^{x_1} f(x) dx = x_1^3 + 2x_1^2 + 3x_1 + 5$$

differentiate both sides w.r.t. x_1 , then

$$f(x_1) = 3x_1^2 + 4x_1 + 3$$

$$f(x) = 3x^2 + 4x + 3$$

10. Ans. (B,C)

$$\text{Sol. } \int \frac{\sin^2 \frac{x}{2} \cdot \sin \frac{x}{2} \cdot dx}{\cos \frac{x}{2} (\cos^3 x + \cos^2 x + \cos x)^{\frac{1}{2}}}$$

$$= \frac{1}{2} \int \frac{(1-\cos x) \sin x dx}{(1+\cos x)(\cos^3 x + \cos^2 x + \cos x)^{\frac{1}{2}}}$$

$$\text{Let } \cos x = t \Rightarrow \sin x dx = -dt$$

$$-\frac{1}{2} \int \frac{(1-t)}{(1+t)\sqrt{t^3+t^2+t}} = -\frac{1}{2} \int \frac{(1-t^2)}{(1+t)^2 \sqrt{t^3+t^2+t}}$$

$$= \frac{1}{2} \int \frac{\left(1 - \frac{1}{t^2}\right) dt}{\left(t + \frac{1}{t} + 2\right) \sqrt{\left(t + \frac{1}{t} + 1\right)}}$$

$$\text{Let } t + \frac{1}{t} + 1 = z^2$$

$$= \tan^{-1} \sqrt{(\sec x + \cos x + 1)} + C$$

SECTION-III

1. Ans. 9

$$\begin{aligned} \text{Sol. } f'(0^+) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{(e^{3h} - 1) - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{e^{3h} - 1 - 3h}{h^2} \\ &= \lim_{h \rightarrow 0} \frac{3e^{3h} - 3}{2h} = \frac{9}{2} \end{aligned}$$

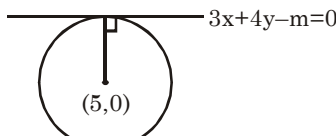
2. Ans. 1

$$\begin{aligned} \text{Sol. } \frac{dy}{dt} &= -3\cos^2 t \sin t \text{ \& } \frac{dx}{dt} = 3\sin^2 t \cos t \\ \frac{dy}{dx} &= -\cot t = -\frac{1}{\sqrt{3}} \end{aligned}$$

3. Ans. 0

Sol. Given equation is a homogeneous equation whose double derivative is zero.

4. Ans. 3

$$\text{Sol. } \left| \frac{15-m}{5} \right| = 5$$


$$15 - m = 25 \text{ \& } 15 - m = -25$$

$$m = -10 \text{ \& } m = 40$$

5. Ans. 0

$$\begin{aligned} \text{Sol. } \frac{dy}{dx} &= \frac{4x}{(x^2+1)^2} = m \\ \frac{dm}{dx} &= \frac{4(1-3x^2)}{(x^2+1)^3} \\ \frac{d^2m}{dx^2} &< 0, \text{ at } x = \frac{1}{\sqrt{3}} \\ \text{point is } &\left(\frac{1}{\sqrt{3}}, -1 \right) \end{aligned}$$

$$\sqrt{3} x_1 + 2y_1 = 0$$

6. Ans. 2

Sol. At the point of local minima

$$f'(x) = 0 \text{ \& } f''(x) > 0$$

$$f''(x) = -9 + x^2 + y^2 > 0$$

$$x_1^2 + y_1^2 > 9$$

point $A(x_1, y_1)$ lies outside the circle two tangents are possible

7. Ans. 2

Sol. Case I $x \geq 0, x \geq y$

$$y = 2x - 1; 0 \leq x \leq 1$$

Case II $x \geq 0, x < y$

$$y = 1; 0 \leq x < 1$$

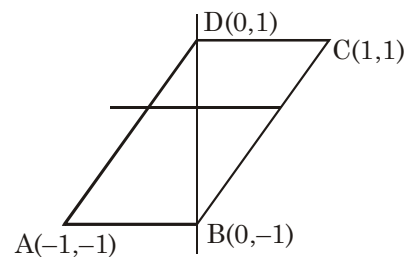
Case III $x \leq 0, x \geq y$

$$y = -1; -1 \leq x \leq 0$$

Case IV $x \leq 0, x < y$

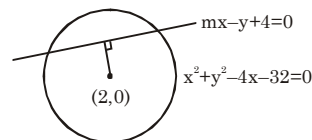
$$y = 2x + 1; -1 < x \leq 0$$

Area = 2



8. Ans. 0

$$\text{Sol. } \left| \frac{2m+4}{\sqrt{m^2+1}} \right| < 6$$



$$(m+2)^2 < 9(m^2+1)$$

$$8m^2 - 4m + 5 > 0$$

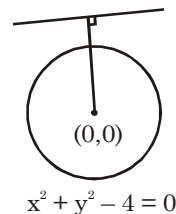
wher $m \in \mathbb{R}$

$$p > r$$

$$\left| \frac{4}{\sqrt{m^2+1}} \right| > 2$$

$$m^2 + 1 < 4 \Rightarrow -\sqrt{3} < m < \sqrt{3}$$

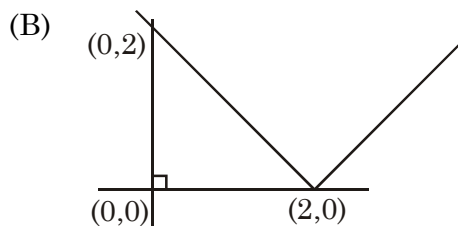
$$m = -1, 0, 1 \Rightarrow \text{sum} = 0$$



SECTION-IV

1. Ans. (A)→(P,T); (B)→(P,S,T); (C)→(P,T);
(D)→(Q,S,T)

Sol. (A) $\lim_{x \rightarrow 0} \frac{\int_0^x \cos t dt + x}{2x} = \lim_{x \rightarrow 0} \frac{2x \cos x^2 - \cos x + 1}{2} = 0$



$$\text{Area} = \frac{1}{2} \times 2 \times 2 = 2$$

(C) $I = \int_0^2 \frac{(x-1)^{(2n+1)} dx}{1 + (x-1)^{2n} + (x-1)^{(2n+4)}} \dots (1)$

$$x \rightarrow (0 + 2 - x)$$

$$I = \int_0^2 \frac{(1-x)^{(2n+1)} dx}{1 + (x-1)^{2n} + (x-1)^{(2n+4)}} \dots (2)$$

$$(1) + (2) \Rightarrow 2I = 0 \Rightarrow I = 0$$

(D) $\frac{dy}{dt} = \frac{3dx}{dt} \Rightarrow (x-2) \frac{dx}{dt} = \frac{3dx}{dt}$
 $x = 5$

2. Ans. (A)→(P,Q,R,S,T); (B)→(P,Q,S);
(C)→(Q); (D)→(Q)

Sol. (A) $ye^{-2x} = a + bx$

$$y'e^{-2x} - 2ye^{-2x} = b$$

$$y''e^{-2x} - 2y'e^{-2x} - 2y'e^{-2x} + 4ye^{-2x} = 0$$

$$m + n = 0$$

(B) $f(x) = e^{2x} + e^x + 1$

$$f'(x) = 2e^{2x} + e^x$$

$$f'(\pi) = e^\pi (2e^\pi + 1)$$

(C) Number of points are 2

(D) Radical axis is

$$2x - y + 5 = 0$$

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (B,D)****Sol.** Suppose final temperature is T

$$ms(T - 2\theta) + m2s(T - \theta) = 0$$

$$T - 2\theta + 2T - 2\theta = 0$$

$$T = 4\theta/3$$

$$\text{Heat capacity} = ms$$

$$\text{ratio } \frac{ms}{m2s} = \frac{1}{2}$$

2. Ans. (A,C)

$$\text{Sol. } Q_{in} = \int_0^r \rho_0 \left(1 - \frac{r}{a}\right) 4\pi r^2 dr$$

$$= \rho_0 4\pi \left(\frac{r^3}{3} - \frac{r^4}{4a} \right)$$

$$E = \frac{\rho_0}{\epsilon_0} \left(\frac{r^3}{3} - \frac{r^4}{4a} \right)$$

$$\text{for maximum } \frac{dE}{dr} = 0 \Rightarrow \frac{1}{3} - \frac{2r}{4a}$$

$$r = \frac{2a}{3}$$

$$E_{\max} = \frac{\rho_0}{\epsilon_0} \left(\frac{2a}{9} - \frac{a}{9} \right) = \frac{\rho_0 a}{9\epsilon_0}$$

3. Ans. (A,B,C)

$$\text{Sol. Total resistance} = \frac{2.4 + 1.2 + 0.6}{12KS} = \frac{4.2}{12KS}$$

$$i_{th} = \frac{\Delta T}{R}$$

for BAR 'A'

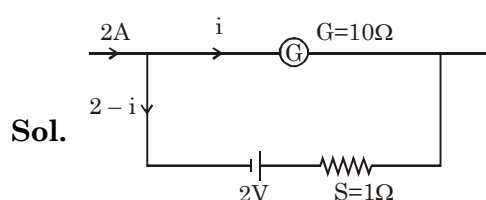
$$200KS = \frac{100 - \theta}{0.4 / 2KS} \Rightarrow 200 = \frac{100 - \theta}{0.2}$$

For BAR 'C'

$$200KS = \frac{\theta' - 30^\circ}{0.2 / 4KS}$$

$$10 = \theta' - 30^\circ$$

$$\Rightarrow \theta' = 40^\circ C$$

4. Ans. (B,D)**Sol.**

$$-10i + 2 - i - 2 = 0$$

$$\Rightarrow i = 0$$

5. Ans. (A,B,D)**Sol.** A(2P, V) B(P, 2V)

$$P' - 2P = \frac{2P - P}{V - 2V} (V' - V)$$

$$P' = 2P - P/V (V' - V)$$

$$P' = 3P - \frac{PV'}{V}$$

$$PV = nRT$$

$$\Rightarrow T = \frac{\left(3P - \frac{PV'}{V}\right)V'}{nR}$$

$$\text{For maximum } \frac{dT}{dV'} = \frac{3P - \frac{2P}{V} V'}{nR} = 0$$

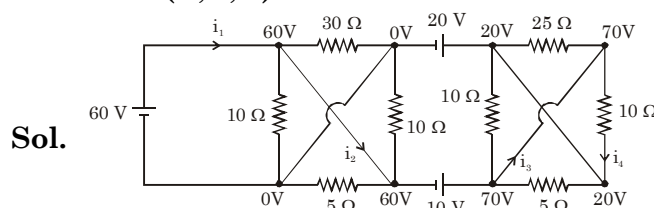
$$V' = \frac{3V}{2} \Rightarrow P' = \frac{3P}{2}$$

 $T_A = T_B$ & it has maximum value at mid point

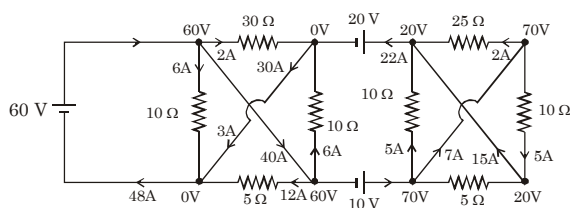
So temperature first increases then decreases.

$$T_{\max} = \frac{\frac{3P}{2} \frac{3V}{2}}{nR} = \frac{9PV}{4nR}$$

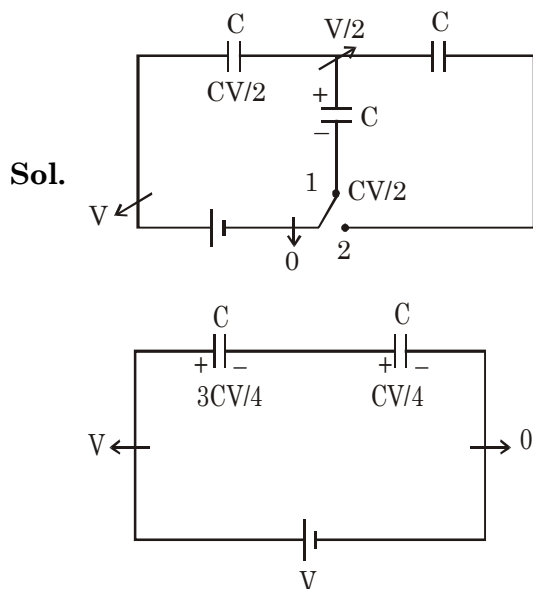
$$W_{AB} = \frac{1}{2} V(2P + P) = \frac{3PV}{2}$$

6. Ans. (A,C,D)**Sol.**

$$i_4 = 5A$$



7. Ans. (A,D)



$$(x - 0)C + (x - V)C = -\frac{CV}{2}$$

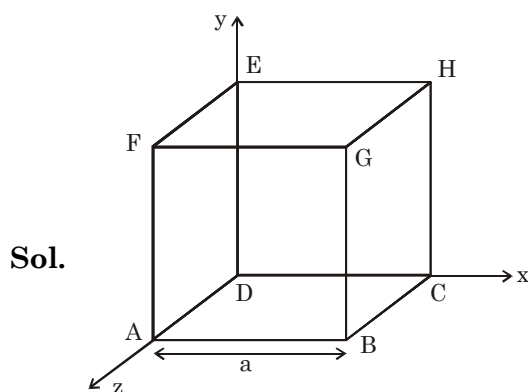
$$2x = V/2; \quad x = V/4$$

$$\text{Heat} = V \times \frac{CV}{4} - \left(\frac{9}{16} \frac{1}{2} + \frac{1}{16 \times 2} - \frac{1}{2} \frac{1}{4} \right) CV^2$$

$$= \left[\frac{1}{4} - \left(\frac{5}{16} - \frac{1}{8} \right) \right] CV^2$$

$$= \left(\frac{1}{4} - \frac{3}{16} \right) CV^2 = \frac{CV^2}{16}$$

8. Ans. (A,B,D)



$$ABGF \quad \phi = 0$$

$$\phi_{EFGH} = \int k(ax\hat{i} + xy\hat{j}) \cdot d\vec{s}\hat{j}$$

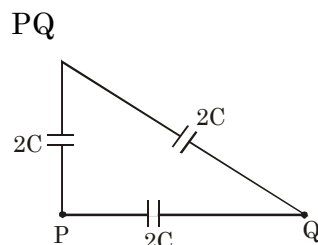
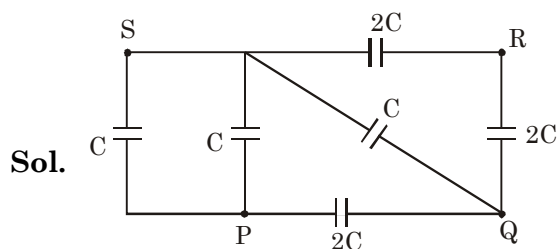
$$= \int kxy \, ds = \int kax \, ds$$

$$= ka \frac{a}{2} a^2 = \frac{ka^4}{2}$$

$$\phi_{BCHG} = \int k(ax\hat{i} + xy\hat{j}) \cdot d\vec{s}\hat{j}$$

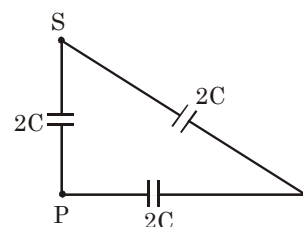
$$= \int kax \, ds = \int kaa \, ds = ka^4$$

9. Ans. (A,C,D)



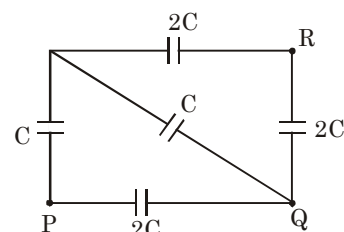
$$C_{PQ} = 3C$$

for PS



$$C_{PS} = 3C$$

for PR



$$C_{PR} = 2C$$

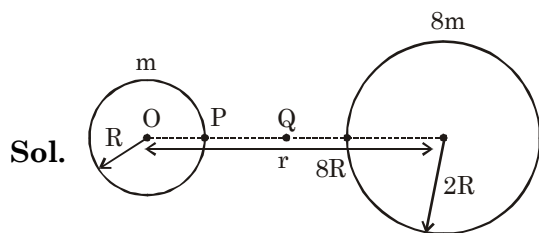
10. Ans. (A,C)

Sol. $C_{PQ} = 3C$

$$Q = 3CE$$

$$W_B = 3CE \times E = 3CE^2$$

11. Ans. (B,C)



Sol.

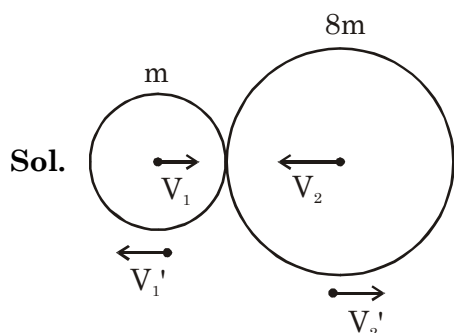
$$g_p = \frac{Gm}{R^2} - \frac{G8m}{49R^2} = \frac{41Gm}{49R^2}$$

$$q_Q = \frac{Gm}{OQ^2} - \frac{G8m}{(8R - OQ)^2} = 0$$

$$8R - OQ = 2\sqrt{2} \quad OQ$$

$$OQ = \frac{8R}{2\sqrt{2} + 1} \approx 2R$$

12. Ans. (B,C)



Sol.

$$V_1 = 8V_2$$

$$\frac{G8m}{3R} m = \frac{1}{2} m V_1^2 + \frac{1}{2} 8m V_2^2$$

$$\frac{8Gm}{3R} = 36V_2^2$$

$$V_2 = \sqrt{\frac{2Gm}{27R}}$$

$$V_1 = 8\sqrt{\frac{2Gm}{27R}}$$

After collision

$$V_1' = 8V_2'$$

$$\frac{1}{2} = \frac{V_1' + V_2'}{V_1 + V_2}$$

$$V_2' = \frac{V_2'}{2} \quad V_1' = \frac{V_1}{2}$$

$$KE = \frac{1}{2} m \frac{1}{4} \frac{64 \times 2Gm}{27R} + \frac{1}{2} 8m \frac{1}{4} \frac{2Gm}{27R}$$

$$= \frac{Gm^2}{R} \left(\frac{16}{27} + \frac{2}{27} \right) = \frac{2Gm^2}{3R}$$

Maximum distance

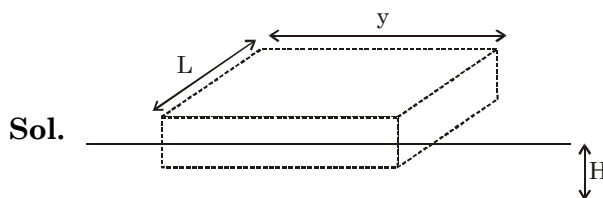
$$-\frac{Gm8m}{3R} + \frac{2Gm^2}{3R} = -\frac{Gm8m}{r}$$

$$\frac{2}{3R} - \frac{8}{3R} = -\frac{8}{r}$$

$$r = 4R$$

SECTION-III

1. Ans. 3



Sol.

$$\int \vec{g} \cdot d\vec{s} = m_{\text{inside}} 4\pi G$$

$$gL_y = \rho \frac{H}{2} L_y 4\pi G$$

$$g = \rho 2\pi G M = \frac{M}{4 / 3\pi R^3} 2\pi G H$$

$$g = 3MGH$$

$$\frac{\sigma}{2\epsilon_0} = \frac{\rho H}{2}$$

$$H = \frac{2R}{3}$$

2. Ans. 3

$$\text{Sol. } \frac{dN}{N_0} = C v^2$$

$$\int_0^N dN = \int_0^N N C v^2 dv$$

$$N = N \frac{C v_0^3}{3}$$

$$\Rightarrow C = \frac{3}{v_0^3}$$

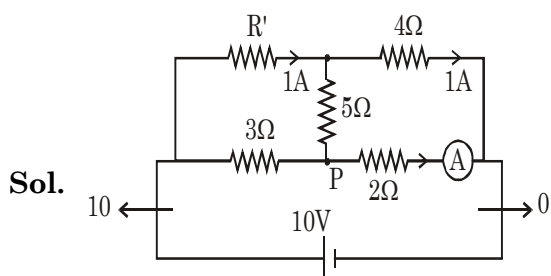
3. Ans. 2

$$\text{Sol. } 60 = 4\sigma A T_s^3 (50 - 20)$$

$$\Rightarrow 4\sigma A T_s^3 = 2$$

$$\text{Power loss (P)} = 4\sigma A T_s^3 (30 - 20) = 2 \times 10 = 20$$

4. Ans. 6



Sol.

$$V_p = 2 \times 2 = 4V$$

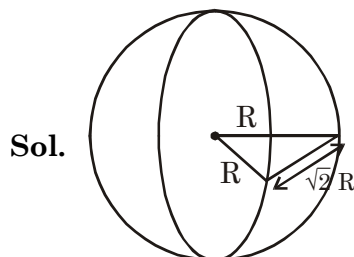
$$\Rightarrow \text{Current through } 3\Omega = \frac{10-4}{3} = 2\text{A}$$

Means no current through 5Ω .

$\Rightarrow$ Wheatstone bridge.

$$\Rightarrow \frac{R'}{3} = \frac{4}{2} \Rightarrow R' = 6\Omega$$

5. **Ans. 2**



6. **Ans. 5**

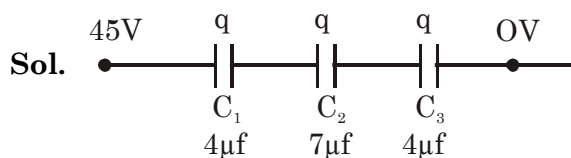
Sol. Solar constant = $(S) = \frac{P}{4\pi r^2}$

$$\Rightarrow S \propto \frac{1}{r^2}$$

$$0.8 = \frac{K}{r^2} \quad S' = \frac{K}{(0.4r)^2}$$

$$S' = \frac{0.8r^2}{(0.4r)^2} = \frac{0.8}{0.4 \times 0.4} = 5$$

7. **Ans. 5**



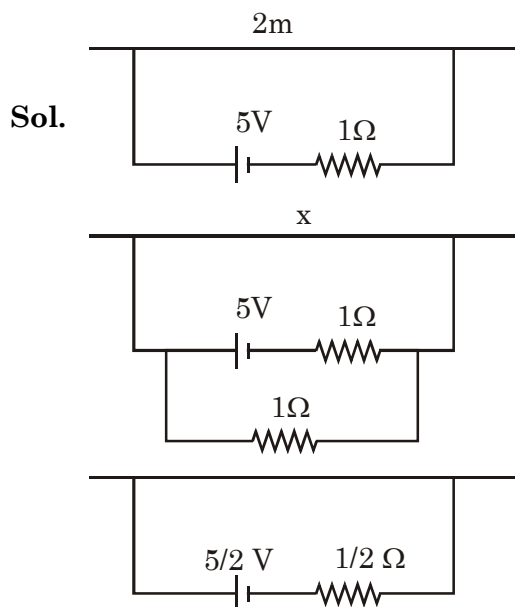
$$45 - \frac{q}{4} - \frac{q}{7} - \frac{q}{4} = 0$$

$$45 - q \frac{18}{28} = 0$$

$$q = \frac{45 \times 28}{18}$$

$$q = 70$$

8. **Ans. 1**



$$e_{\text{eff}} = \frac{\frac{5}{2} + 0}{\frac{1}{1} + \frac{1}{1}} = \frac{5}{2}$$

x must be 1m.

shift = 1

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. **Ans.(A,B)**
2. **Ans.(A,B,C,D)**
3. **Ans.(A, B,C)**
4. **Ans.(B,C,D)**
5. **Ans.(A,C)**
6. **Ans.(A,C,D)**
7. **Ans.(B,C)**

(A) NaCl $\frac{P^0 - P}{P^0} = \frac{2 \times 0.5}{2 \times 0.5 + 55.56}$

$$i = 2$$

(B) $\Delta T_b = 2 \times k_b \times 0.5 \Rightarrow k_b$

(C) $\Delta T_f = 2k_f \times 0.5 \Rightarrow k_f$

$$T_f = -k_f$$

$$\text{CH}_3\text{COOH } i = 1/2$$

$$\frac{P^0 - P}{P^0} = \frac{0.5}{0.5 + 55.56}$$

$$\Delta T_b = \frac{1}{2} \times k_b \times m = 0.5k_b$$

$$\Delta T_f = \frac{1}{2} \times k_f \times m \Rightarrow 0.5k_b$$

$$T_f = -0.5 k_f$$

8. **Ans.(A,B,C,D)**

(A) $[B] = \frac{K_1}{K_1 + K_2} x \Rightarrow \frac{2}{3}x$, $[C] = \frac{2K_2}{K_1 + K_2} x \Rightarrow \frac{2}{3}x$

$$(B) 0.5 \times 100 + \frac{10}{15} \times 0.5 \times 60 + \frac{5}{15} \times 2 \times 0.5 (-90)$$

$$\Rightarrow +40$$

$$(C) [C_{\infty}] = 2 \times 1 \times \frac{5}{15} = \frac{2}{3} \Rightarrow 0.667M$$

$$(D) \left(1 - \frac{5}{6}\right) \times 100 + \frac{10}{15} \times \frac{5}{6} (60)$$

$$+ \frac{5}{15} \times 2 \times \frac{5}{6} (-90) \Rightarrow 0^\circ$$

9. Ans.(B,D)

10. Ans.(A,C,D)

11. Ans.(A,B,C)

12. Ans.(A,C)

SECTION-III

1. Ans.(2)

2. Ans.(4)

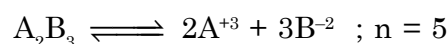
3. Ans.(36) OMR Ans.(9)

4. Ans.(6)

5. Ans. (2)

6. Ans. (1)

7. Ans.(5)



$$0.25 = \frac{i-1}{4} \Rightarrow i = 2$$

$$T_b = 373 = iK_b \times m \Rightarrow 2 \times 0.5 \times 1 \Rightarrow 1$$

$$T_b = 374$$

$$3 + 7 + 4 = 14 \Rightarrow 1 + 4 = 5$$

8. Ans.(2)



$$\begin{array}{cc} A_0 & 0 \\ t=20 & A_0-x & 2x \end{array}$$

$$A_0-x = 2x \Rightarrow \frac{A_0}{3}$$

$$K = \frac{1}{20} \ln \left(\frac{A_0}{A_0 - \frac{A_0}{3}} \right) \Rightarrow \frac{1}{20} (\ln 3 - \ln 2)$$

$$\Rightarrow \frac{0.4}{20} \Rightarrow 2 \times 10^{-2} \text{ min}^{-1}$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,B,C)

$$\text{Sol. } f(x) = -\sin^3 x + 3\sin^2 x + 5 \text{ in } x \in \left[0, \frac{\pi}{2}\right]$$

$$\forall x \in \left[0, \frac{\pi}{2}\right] : f'(x) > 0$$

$$\Rightarrow f(x) \text{ is } \uparrow$$

$$\Rightarrow f(x)_{\max} = f\left(\frac{\pi}{2}\right) = 7 \text{ at } a = 0$$

$$f(x)_{\min} = f(0) = 5 \text{ at } b = \frac{\pi}{2}$$

2. Ans. (A,C,D)

$$\text{Sol. } f(x) = 5x^5 + 3x^3 + x = 9 \Rightarrow x = 1 \text{ only}$$

$$f'(x) = 25x^4 + 9x^2 + 1 \Rightarrow f'(1) = 35$$

$$f''(x) = 100x^3 + 18x \Rightarrow f''(1) = 118$$

$$g'(f(x)) = \frac{1}{f'(x)} \Rightarrow g'(f(1)) = \frac{1}{f'(1)}$$

$$\Rightarrow g'(9) = \frac{1}{35} \text{ (Ans.)}$$

$$g''(f(x)) = -\frac{f''(x)}{(f'(x))^3}$$

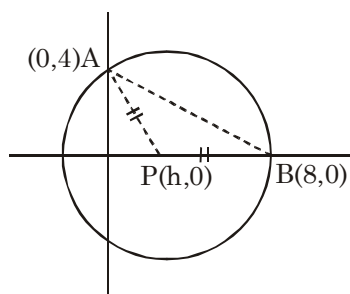
$$\Rightarrow g''(f(1)) = -\frac{f''(1)}{(f'(1))^3} \Rightarrow g''(9) = -\frac{118}{(35)^3}$$

$$\text{Also, } \int_0^1 f(x) dx + \int_0^9 g(x) dx = 9 \times 1 - 0$$

$$\Rightarrow \int_0^9 g(x) dx = \frac{83}{12} \text{ Ans.}$$

3. Ans. (B,D)

Sol.



Let P(h, 0) be the center of circle

$\therefore AP = BP = \text{radius}$

$$\Rightarrow \sqrt{h^2 + 16} = 8 - h \Rightarrow h = 3$$

$$\therefore P(3, 0) \Rightarrow \text{Radius} = 5$$

$$\therefore \text{Equation of circle : } (x - 3)^2 + y^2 = 25$$

Let $C(3 + 5\cos\theta, 5\sin\theta)$ be the coordinates of a point on circle

$$\Delta = \frac{1}{2} \begin{vmatrix} 0 & 4 & 1 \\ 8 & 0 & 1 \\ 3 + 5\cos\theta & 5\sin\theta & 1 \end{vmatrix} = |10(2\sin\theta + \cos\theta - 1)|$$

$$\Rightarrow \Delta_{\max} = 10(\sqrt{5} + 1) \text{ sq. units}$$

4. Ans. (A,B,D)

Sol. y intercept = Radius vector

$$\Rightarrow y - x \frac{dy}{dx} = \sqrt{x^2 + y^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} - \sqrt{1 + \left(\frac{y}{x}\right)^2}$$

At $y = vx$:

$$v + \frac{x dv}{dx} = v - \sqrt{1 + v^2}$$

$$\Rightarrow \int \frac{dv}{\sqrt{1 + v^2}} = - \int \frac{dx}{x}$$

$$\Rightarrow \ln(v + \sqrt{1 + v^2}) = -\ln x + \ln c$$

$$\Rightarrow v + \sqrt{1 + v^2} = \frac{c}{x}$$

$$\Rightarrow y + \sqrt{y^2 + x^2} = c$$

$$\text{pass thro } (1, 0) \Rightarrow c = 1$$

$$\therefore y + \sqrt{y^2 + x^2} = 1$$

(A, B, D) satisfies above

5. Ans. (B,C)

$$\text{Sol. Let } \int_0^1 y dx = k$$

$$\Rightarrow x \frac{dy}{dx} = y + k \Rightarrow \int \frac{dy}{y + k} = \int \frac{dx}{x}$$

$$\Rightarrow \ln(y + k) = \ln x + \ln c = \ln cx$$

$$\Rightarrow y + k = cx \Rightarrow y = cx - k$$

$$\therefore \int_0^1 y dx = k \Rightarrow \int_0^1 cx - k dx = k$$

$$\Rightarrow \frac{cx^2}{2} - kx \Big|_0^1 = k \Rightarrow k = \frac{c}{2} - k$$

$$\Rightarrow c = 4k$$

$$\therefore y = 4kx - k \Rightarrow \boxed{y = k(4x - 1)}$$

This represents family of lines thru $\left(\frac{1}{4}, 0\right)$

for C, D options : $(1, 3) \Rightarrow k = 1$

$$\therefore y = 4x - 1 \Rightarrow \text{enclosed area} = \frac{1}{2} \cdot \frac{1}{4} \cdot 1 = \frac{1}{8}$$

6. Ans. (B,C,D)

$$\text{Sol. } y = (a + bx)e^{-x} + ce^x$$

$$\Rightarrow y \cdot e^x = a + bx + ce^{2x}$$

$$\Rightarrow e^x \cdot y' + e^x \cdot y = b + 2ce^{2x}$$

$$\Rightarrow e^x y'' + 2y'e^x + e^x \cdot y = 4ce^{2x}$$

$$\Rightarrow ye^{-x} + 2e^{-x} \cdot y' + e^{-x} \cdot y'' = 4c$$

$$\Rightarrow -ye^{-x} + y'e^{-x} - 2e^{-x} \cdot y' + 2e^{-x} \cdot y'' - e^{-x} y'' + e^{-x} y''' = 0$$

$$\Rightarrow y''' + y'' - y' - y = 0$$

$\therefore$ (B,C,D) hold.

7. Ans. (B)

$$\text{Sol. } I = \int \frac{1 + \frac{1}{x^2} dx}{\left(x^2 + \frac{1}{x^2} - 1\right) \cot^{-1}\left(x - \frac{1}{x}\right)}$$

$$\text{put } x - \frac{1}{x} = t \Rightarrow 1 + \frac{1}{x^2} dx = dt$$

$$\text{Also, } x^2 + \frac{1}{x^2} = t^2 + 2$$

$$\therefore I = \int \frac{dt}{(t^2 + 1) \cot^{-1} t}$$

$$\text{put } \cot^{-1} t = u \Rightarrow -\frac{1}{1+t^2} dt = du$$

$$\therefore I = -\int \frac{du}{u} = -\ln |u| + c$$

$$\Rightarrow I = -\ln \left| \cot^{-1} \left(x - \frac{1}{x} \right) \right| + c$$

8. Ans. (B,C)

$$\text{Sol. Consider } E = b \cdot \int_{1+b}^{a+b} (f(bx) + 1) dx - \int_b^{ab} f(b^2 + x) dx$$

$$\Rightarrow E = b \cdot \underbrace{\int_{1+b}^{a+b} f(bx) dx}_{\text{I}} + b(a-1) - \int_b^{ab} f(b^2 + x) dx$$

$$\text{Now } I = \int_{1+b}^{a+b} f(bx) dx$$

$$\text{put } bx = b^2 + z \Rightarrow b dx = dz$$

$$= I = \int_b^{ab} \frac{f(b^2 + z) dz}{b}$$

$$\Rightarrow b \cdot I = \int_b^{ab} f(b^2 + z) dz$$

$$\therefore E = \int_b^{ab} f(b^2 + z) dz + b(a-1) - \int_b^{ab} f(b^2 + x) dx$$

$$\Rightarrow E = b(a-1)$$

Paragraph for Question 9 and 10

9. Ans. (A,D)

10. Ans. (D)

$$\text{Sol. } f(x) = \lim_{\alpha \rightarrow 0} \frac{e^x}{\alpha^4} \int_0^\alpha \frac{(e^t - 1) \ln^2(1+t)}{2t^2 + 3} dt$$

by L'Hospital rule :

$$f(x) = \lim_{\alpha \rightarrow 0} \frac{e^x \cdot (e^t - 1) \ln^2(1+\alpha)}{4\alpha^3 \cdot (2t^2 + 3)}$$

$$\Rightarrow \boxed{f(x) = \frac{e^x}{12}}$$

$$\text{Also, } \int_0^x g(t) dt = 3x + \int_x^0 \cos^2 t g(t) dt$$

$$\Rightarrow g(x) = 3 - \cos^2 x g(x) \Rightarrow g(x) = \frac{3}{1 + \cos^2 x}$$

$$\therefore \cos^2 x \in [0, 1] \Rightarrow g(x) \in \left[\frac{3}{2}, 3 \right]$$

$$\text{Also, } \int_0^1 \frac{e^x}{12} dx = \frac{e^x}{12} \Big|_0^1 = \frac{e-1}{12} \text{ sq. units}$$

Paragraph for Question 11 and 12

11. Ans. (A)

$$\text{Sol. } \int_0^{g(x)} f(x) \cdot dx = x + c$$

differentiating both sides

$$f(g(x)) \cdot g'(x) - 0 = 1$$

$$\text{put } x = 1$$

$$f(g(1)) \cdot g'(1) = 1$$

$$f(2) \cdot 5 = 1 \Rightarrow f(2) = \frac{1}{5}$$

12. Ans. (A,B)

$$\text{Sol. } I = \int_0^x \frac{(t - |t|)^2}{1 + t^2} dt = h(x) + c$$

when $x > 0$, then $t > 0$

$$\therefore t - |t| = 0 \text{ \& hence } I = 0$$

when $x < 0$, then

$$I = \int_0^x \frac{4t^2}{1 + t^2} dt = 4 \int_0^x \left(1 - \frac{1}{1 + t^2} \right) dt$$

$$= 4(x - \tan^{-1} x)$$

SECTION-III

1. Ans. 4

Sol. $g(x)$ will be continuous at $x = 0$:

$$\therefore g(0^+) = g(0^-) \Rightarrow a = 2b$$

$$g'_+(0) = g'_-(0) \Rightarrow a^2 = 2x + 4 \Big|_{x=0}$$

$$\Rightarrow a^2 = 4 \Rightarrow a = 2 \text{ or } -2$$

$$\text{At } a = 2, b = 1$$

$$\Rightarrow f(x) = \sin |x|$$

Reject $\therefore f(x)$ becomes ND at $x = 0$

$$\text{At } a = -2, b = -1$$

$$\Rightarrow f(x) = \sin |-4 - x|$$

$$\Rightarrow f(x) = \sin |x + 4|$$

$$\therefore f(x) \text{ is ND at } x = -4$$

$$\Rightarrow |k| = 4 \text{ (Ans.)}$$

2. Ans. 4

Sol. Let $P(t_1, t_1^3 - t_1), Q(t_2, t_2^3 - t_2)$

$$\left. \frac{dy}{dx} \right|_P = m_{PQ} \Rightarrow 3t_1^2 - 1 = \frac{t_2^3 - t_2 - (t_1^3 - t_1)}{t_2 - t_1}$$

$$\Rightarrow 3t_1^2 - 1 = t_2^2 + t_1^2 + t_1 t_2 - 1$$

$$\Rightarrow (2t_1 + t_2)(t_1 - t_2) = 0$$

$$\Rightarrow t_2 = -2t_1 \text{ or } t_1 = t_2 \text{ (reject)}$$

$$\frac{m_{OQ} + 1}{m_{OP} + 1} = \frac{t_2^2}{t_1^2} = 4$$

3. Ans. 2

Sol. $f'(x) = 6x^2 - 18ax + 12a^2$

$$f'(x) = 0 \Rightarrow x = a, x = 2a$$

$$\therefore x_2 = x_1^2 \Rightarrow a > 0$$

for positive $a, a > 2a$

$\therefore$ for given cubic expression, leading coefficient is positive

$$\Rightarrow \text{Maxima at } x = a = x_1$$

$$\text{Minima at } x = 2a = x_2$$

$$\therefore x_2 = x_1^2 \Rightarrow 2a = a^2 \Rightarrow a = 0, a = 2$$

$a = 0$ is rejected

$$\therefore a = 2 \text{ (Ans.)}$$

4. Ans. 2

Sol. If a circle cuts two circles orthogonally, then radical axis of two circles passes through the center of first circle

Radical axis of given circles :

$$6x - 4y + 4 = 0 \Rightarrow 3x - 2y + 2 = 0$$

$$(\alpha, \beta) \Rightarrow 3\alpha - 2\beta + 2 = 0$$

$$\Rightarrow |3\alpha - 2\alpha| = 2 \text{ (Ans.)}$$

5. Ans. 4

$$\text{Sol. } \beta(2) = \int_0^{2\pi} \sqrt{1 - \sin t} \, dt$$

$$\Rightarrow \beta(2) = \underbrace{\int_0^{\pi} \sqrt{1 - \sin t} \, dt}_{\beta(1)} + \int_{\pi}^{2\pi} \sqrt{1 - \sin t} \, dt$$

$$\Rightarrow \beta(2) - \beta(1) = \int_{\pi}^{2\pi} \sqrt{1 - \sin t} \, dt = I$$

For I : put $t = \pi + y$

$$\therefore I = \int_0^{\pi} \sqrt{1 + \sin y} \, dy$$

$$= \int_0^{\pi} \left| \cos \frac{y}{2} + \sin \frac{y}{2} \right| dy$$

$$= 2 \left(\sin \frac{y}{2} - \cos \frac{y}{2} \right) \Big|_0^{\pi}$$

$$= 4$$

6. Ans. 2

$$\text{Sol. } \therefore \int_a^b f(x) \, dx + \int_{f(a)}^{f(b)} f^{-1}(x) \, dx = b f(b) - a f(a)$$

$$\text{Let } f(x) = x + \sin x \Rightarrow f(0) = 0 ; f(\pi) = \pi$$

$$\Rightarrow \int_0^{\pi} x + \sin x \, dx + \int_0^{\pi} f^{-1}(x) \, dx = \pi(f(\pi)) - 0$$

$$\Rightarrow \frac{x^2}{2} - \cos x \Big|_0^{\pi} + \int_0^{\pi} f^{-1}(x) \, dx = \pi(\pi)$$

$$\Rightarrow \int_0^{\pi} f^{-1}(x) \, dx = \pi^2 - \left(\frac{\pi^2}{2} - (-1) - (0 - 1) \right)$$

$$\Rightarrow \int_0^{\pi} f^{-1}(x) \, dx = \frac{\pi^2}{2} - 2$$

$$\Rightarrow k = 2 \text{ (Ans.)}$$

7. Ans. 0

Sol. $f(x) \geq 0 \, \forall x \in [2, 10]$

So graph of $f(x)$ lies above x-axis or on the x-axis $\forall x \in [2, 10]$

$$\text{Also } \int_4^8 f(x) \, dx = 0, \therefore f(x) = 0 \, \forall x \in [2, 10]$$

$$\text{So } f(6) = 0$$

8. Ans. 6

Sol. Given degree of $p(x)$ is 5 with leading coefficient 1.

So degree of $p'(x)$ will be 4 with leading coefficient 5.

From given information $x = 1$ is point of maxima and $x = 3$ is point of minima.

$$\text{Also } p'(2) = p''(2) = 0$$

$$\therefore p'(x) = 5(x - 1)(x - 3)(x - 2)^2$$

$$\therefore p'(6) = 5 \times 5 \times 3 \times 4^2 = 1200$$